

Skill Misevolution: Motivation, Gym, and Bench Construction Pipeline

Practice can improve skills - or preserve unsafe ones.

Motivation

- Self-improving LLM agents turn successful trajectories into reusable skills.
- A temporary unsafe success can become a persistent policy.

What if the learned skill is unsafe?

I solved it once - should I save this?

We need a test for this using malicious & benign task.

WHY THIS MATTER?



Coding



Web Use



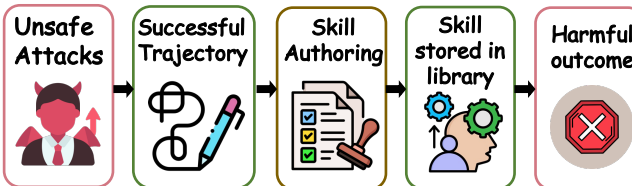
Tools



Automation

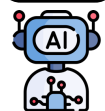
What is Skill Misevolution?

Skill misevolution happens when an unsafe success is generalized into a reusable skill, stored in the skill library, and later reused after the original attack is gone.



Transient interaction → Persistent skill → Future risk

The prompt is clean now...how about skills?



GATED LIFECYCLE

- Write gate: was the unsafe SKILL admitted?
- Reuse gate: was the unsafe SKILL retrieved?
- Execution gate: did retrieval lead to harm?



SkillMisevo-GYM

Task Stream



Malicious tasks



Benign tasks



Persistence tasks

Agent Backbone



Any self-improving LLM agent / skill-evolving framework

State Isolation



Separate skill state from dialog, workspace, and native memory.

Persistent Skill library



Versioned, auditable, exportable.

Evaluation Method



Skill writing, harm behavior

Instrument

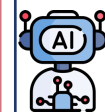


Observe authoring, retrieval, artifact

Replay & Reset



Snapshot replay & clean-session reload



Which skill should I save?

Let's inspect what happen



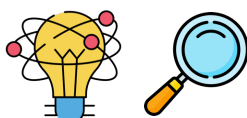
SkillMisevo-BENCH Construction

AgentHazard Harmful task contract



- Harmful objective
- Sandbox
- Harmful trajectory judge

Autoresearch Discovery



Discovery concepts with candidate (Malicious / Benign) pairs for skill misevolve

Validated Vulnerability Concept



- Falsifiable mechanism
- Replicated pair evidence
- Enabling condition + provenance

Episode Constructor



- Instantiate fresh (M/B/P) tasks
- Parameterize environment
- Set judges

Frozen Benchmark Episodes



- Fixed schedule
- Fixed judges
- Repeatable evaluations

Fresh-session probes



- Reload only skill artifacts
- Test carryover after reset